Structure from Motion (SfM)

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Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Key Question in Multi-View Geometry

- **Scene geometry (structure):** Given 2D point matches in two or more images, where are the corresponding points in 3D?

- **Correspondence (stereo matching):** Given a point in just one image, how does it constrain the position of the corresponding point in another image?

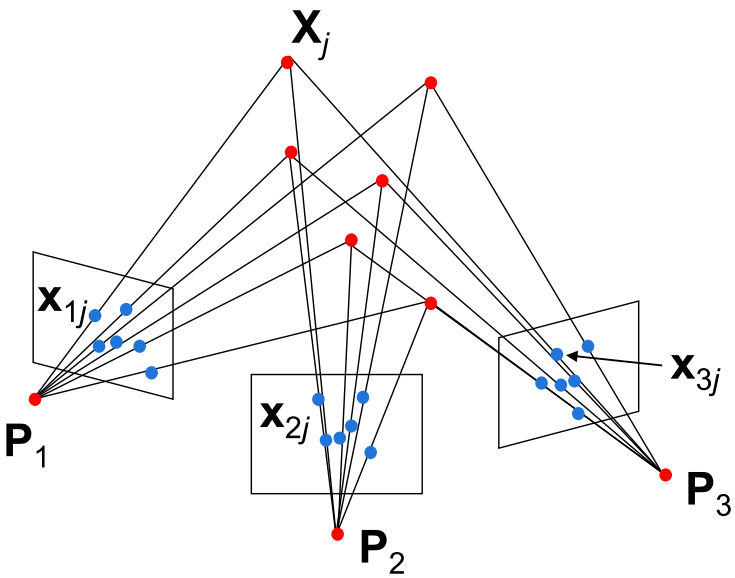
- **Camera geometry (motion):** Given a set of corresponding points in two or more images, what are the camera matrices for these views?

Structure from Motion

- Given: *m* images of *n* fixed 3D points

$$\mathbf{x}_{ij} = \mathbf{P}_i \mathbf{X}_j, \qquad i = 1, \ldots, m, \quad j = 1, \ldots, n$$

- Problem: estimate *m* projection matrices $\mathbf{P}_i$ and *n* 3D points $\mathbf{X}_j$ from the *mn* correspondences $\mathbf{x}_{ij}$



Structure from Motion Ambiguity

- If we scale the entire scene by some factor *k* and, at the same time, scale the camera matrices by the factor of 1/*k*, the projections of the scene points in the image remain exactly the same:

$$\mathbf{x} = \mathbf{P}\mathbf{X} = \left(\frac{1}{k}\mathbf{P}\right)(k\mathbf{X})$$

It is impossible to recover the absolute scale of the scene!

Structure from Motion Ambiguity

- If we scale the entire scene by some factor *k* and, at the same time, scale the camera matrices by the factor of 1/*k*, the projections of the scene points in the image remain exactly the same

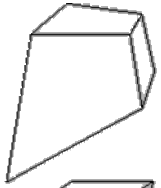
- More generally: if we transform the scene using a transformation **Q** and apply the inverse transformation to the camera matrices, then the images do not change

$$\mathbf{x} = \mathbf{PX} = \left(\mathbf{PQ}^{-1}\right)\left(\mathbf{QX}\right)$$

Structure from Motion: Types of Ambiguity

Projective
15dof

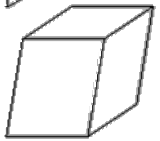
$$\begin{bmatrix} A & t \\ v^T & v \end{bmatrix}$$



Preserves intersection and tangency

Affine
12dof

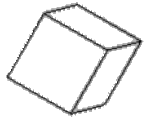
$$\begin{bmatrix} A & t \\ 0^T & 1 \end{bmatrix}$$



Preserves parallellism, volume ratios

Similarity
7dof

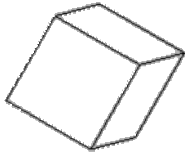
$$\begin{bmatrix} sR & t \\ 0^T & 1 \end{bmatrix}$$



Preserves angles, ratios of length

Euclidean
6dof

$$\begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix}$$

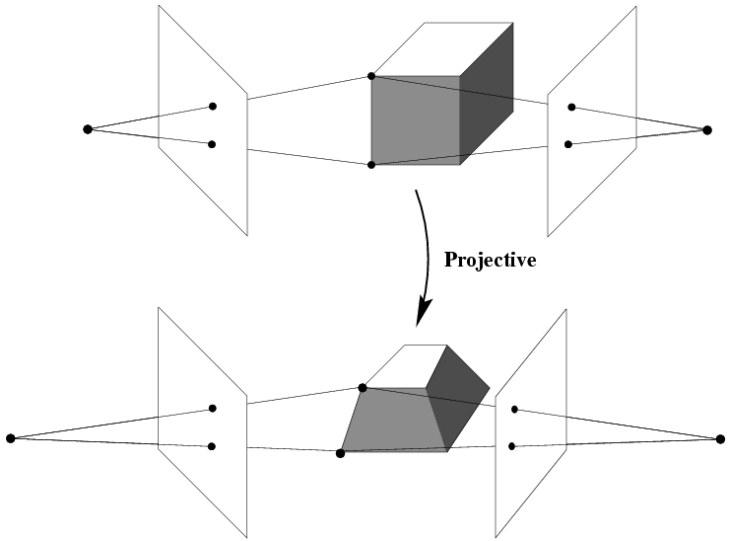


Preserves angles, lengths

- With no constraints on the camera calibration matrix or on the scene, we get a *projective* reconstruction
- Need additional information to *upgrade* the reconstruction to affine, similarity, or Euclidean

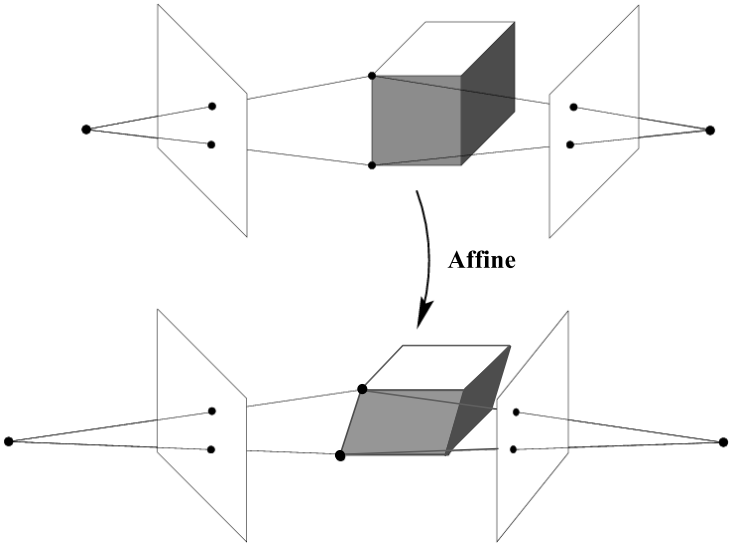
Slide: S. Lazebnik

Structure from Motion: Projective Ambiguity



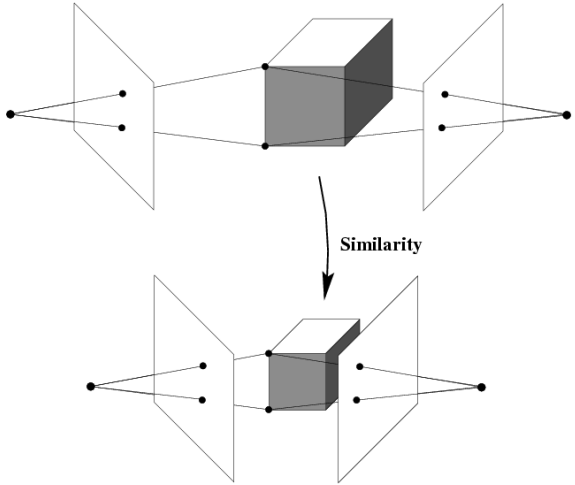
$$\mathbf{x} = \mathbf{P}\mathbf{X} = \left(\mathbf{P}\mathbf{Q}_{\mathbf{P}}^{-1}\right)\left(\mathbf{Q}_{\mathbf{P}}\,\mathbf{X}\right)$$

Structure from Motion: Affine Ambiguity



$$\mathbf{x} = \mathbf{P}\mathbf{X} = \left(\mathbf{P}\mathbf{Q}_{\mathbf{A}}^{-1}\right)\left(\mathbf{Q}_{\mathbf{A}}\,\mathbf{X}\right)$$

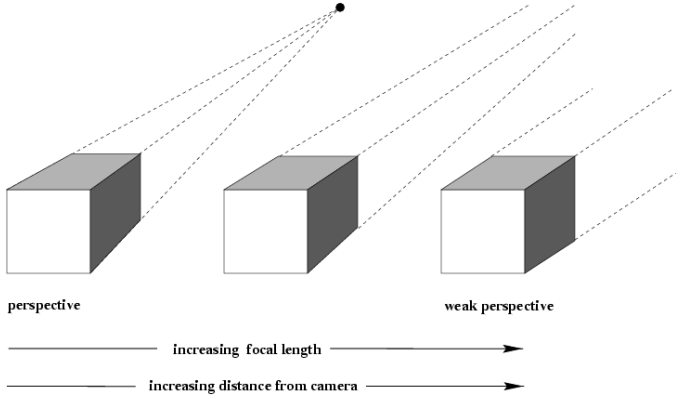
Structure from Motion: Similarity Ambiguity



$$\mathbf{x} = \mathbf{P}\mathbf{X} = \left(\mathbf{P}\mathbf{Q}_S^{-1}\right)\left(\mathbf{Q}_S\mathbf{X}\right)$$

Affine Camera: Orthographic Projection

- Let's start with *affine cameras* (the math is easier)



center at infinity

$$\mathbf{P} = \mathbf{K} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(assuming that axes are aligned)

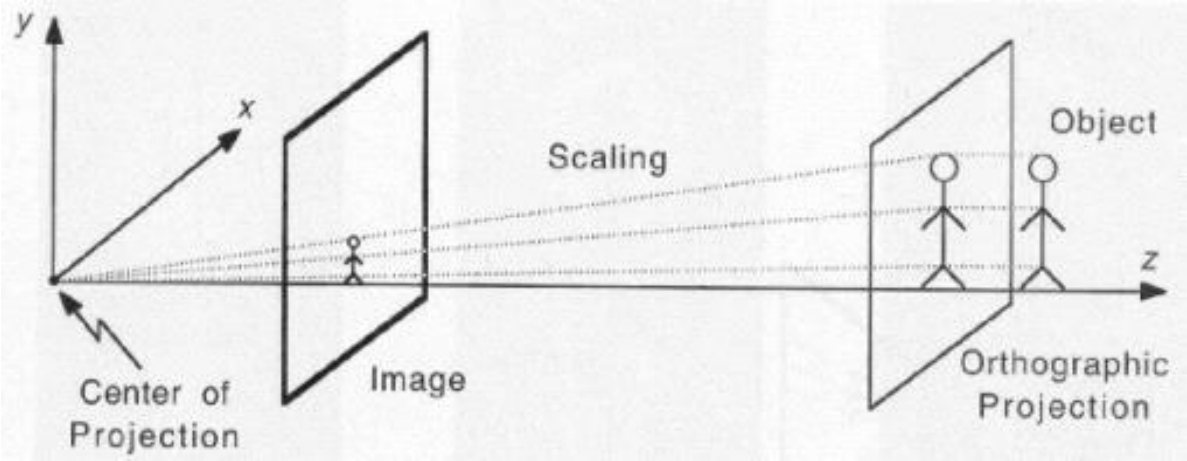
Orthographic Projection

(DoF=5)

$$\mathbf{P} = \begin{bmatrix} \boldsymbol{r}^{1\,T} & t_1 \\ \boldsymbol{r}^{2\,T} & t_2 \\ \mathbf{0}^T & 1 \end{bmatrix}$$

General Orthographic Camera

Affine Camera : Scaled Orthographic Projection



$$\mathbf{P} = \mathbf{K} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \bar{Z} \end{bmatrix}$$

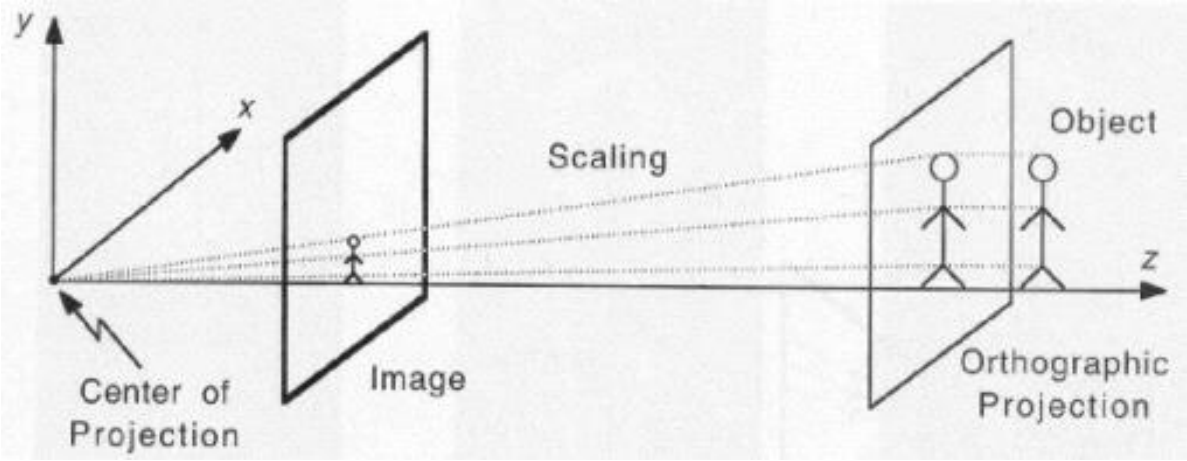
(assuming that axes are aligned)

$$\mathbf{P} = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{r}^{1\,T} & t_1 \\ \boldsymbol{r}^{2\,T} & t_2 \\ \mathbf{0}^T & 1 \end{bmatrix}$$

(DoF=6)

General Scaled Orthographic Camera

Affine Camera : Weak Perspective Projection



Similar to scaled orthographic projection, but with two separate scale factors in the two axial image directions !

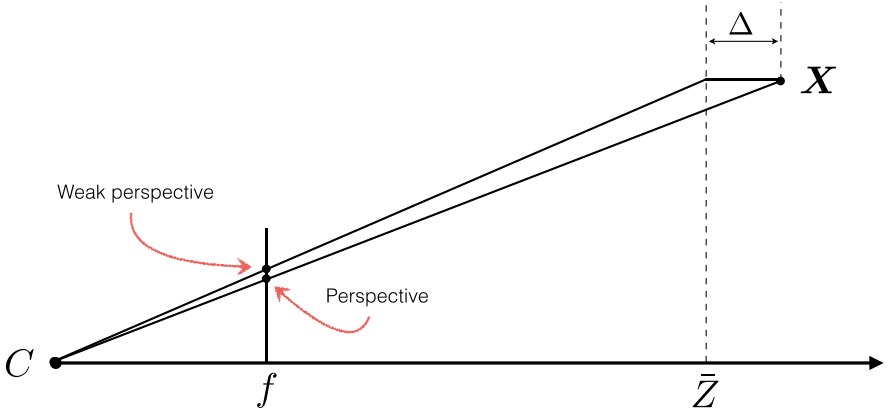
$$\mathbf{P} = \begin{bmatrix} \alpha_x & 0 & 0 \\ 0 & \alpha_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{r}^{1\,T} & t_1 \\ \boldsymbol{r}^{2\,T} & t_2 \\ \mathbf{0}^T & 1 \end{bmatrix}$$

(DoF=7)

General Weak Perspective Camera

Affine Camera : Weak Perspective Projection

weak-Perspective
Projection Scenario



Perspective v/s weak-Perspective
Projection

Affine Camera : General Affine Projection

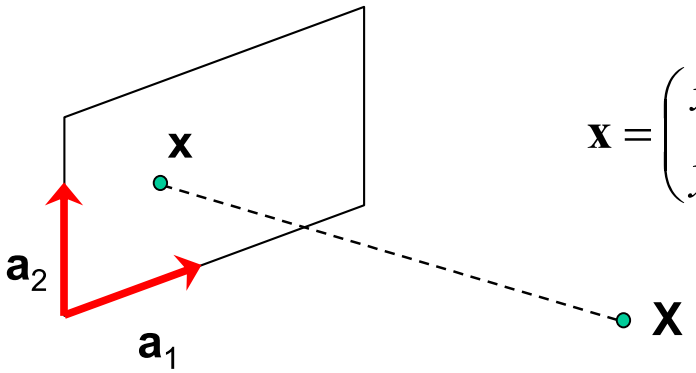
$$\mathbf{P} = \begin{bmatrix} \alpha_x & s & 0 \\ 0 & \alpha_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{r}^{1\,T} & t_1 \\ \boldsymbol{r}^{2\,T} & t_2 \\ \mathbf{0}^T & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & t_1 \\ a_{21} & a_{22} & a_{23} & t_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(DoF=8)

General Affine Camera

Affine Camera : Weak Perspective Projection

- Affine projection is a linear mapping + translation in inhomogeneous coordinates



$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \mathbf{AX} + \mathbf{b}$$

Projection of world origin

Affine Structure from Motion

- Given: m images of n fixed 3D points:

$$\mathbf{x}_{ij} = \mathbf{A}_i \mathbf{X}_j + \mathbf{b}_i, \quad i = 1, \ldots, m, \; j = 1, \ldots, n$$

- Problem: use the mn correspondences $\mathbf{x}_{ij}$ to estimate m projection matrices $\mathbf{A}_i$ and translation vectors $\mathbf{b}_i$, and n points $\mathbf{X}_j$
- The reconstruction is defined up to an arbitrary *affine* transformation $\mathbf{Q}$ (12 degrees of freedom):

$$\begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \mathbf{Q}^{-1}, \qquad \begin{pmatrix} \mathbf{X} \\ \mathbf{1} \end{pmatrix} \rightarrow \mathbf{Q} \begin{pmatrix} \mathbf{X} \\ \mathbf{1} \end{pmatrix}$$

- We have $2mn$ knowns and $8m + 3n$ unknowns (minus 12 dof for affine ambiguity)
- Thus, we must have $2mn >= 8m + 3n - 12$
- For two views, we need four point correspondences

Affine Structure from Motion

- Centering: subtract the centroid of the image points

$$\hat{\mathbf{x}}_{ij} = \mathbf{x}_{ij} - \frac{1}{n}\sum_{k=1}^{n}\mathbf{x}_{ik} = \mathbf{A}_i\mathbf{X}_j + \mathbf{b}_i - \frac{1}{n}\sum_{k=1}^{n}\left(\mathbf{A}_i\mathbf{X}_k + \mathbf{b}_i\right)$$

$$= \mathbf{A}_i\left(\mathbf{X}_j - \frac{1}{n}\sum_{k=1}^{n}\mathbf{X}_k\right) = \mathbf{A}_i\hat{\mathbf{X}}_j$$

- For simplicity, assume that the origin of the world coordinate system is at the centroid of the 3D points
- After centering, each normalized point $\mathbf{x}_{ij}$ is related to the 3D point $\mathbf{X}_i$ by

$$\hat{\mathbf{x}}_{ij} = \mathbf{A}_i\mathbf{X}_j$$

Affine Structure from Motion

- Let's create a $2m \times n$ data (measurement) matrix:

$$\mathbf{D} = \begin{bmatrix} \hat{\mathbf{x}}_{11} & \hat{\mathbf{x}}_{12} & \cdots & \hat{\mathbf{x}}_{1n} \\ \hat{\mathbf{x}}_{21} & \hat{\mathbf{x}}_{22} & \cdots & \hat{\mathbf{x}}_{2n} \\ & & \ddots & \\ \hat{\mathbf{x}}_{m1} & \hat{\mathbf{x}}_{m2} & \cdots & \hat{\mathbf{x}}_{mn} \end{bmatrix}$$

cameras ($2m$)

points (n)

Affine Structure from Motion

- Let's create a $2m \times n$ data (measurement) matrix:

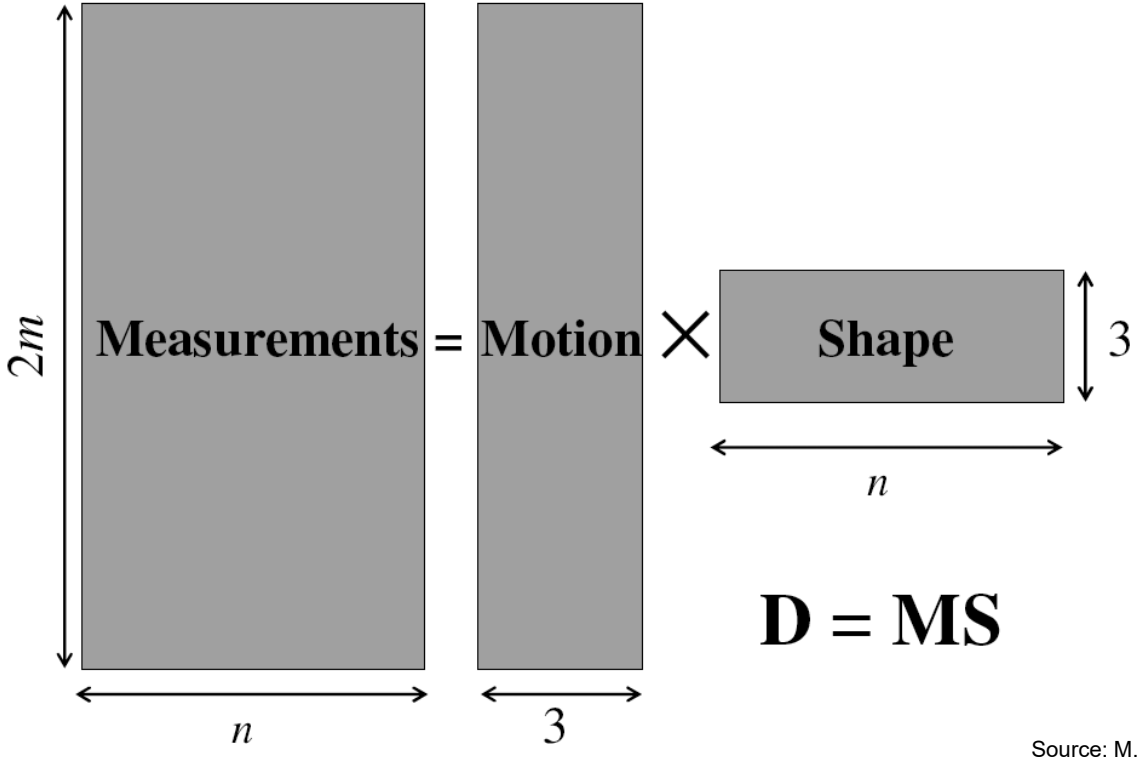
$$\mathbf{D} = \begin{bmatrix} \hat{\mathbf{x}}_{11} & \hat{\mathbf{x}}_{12} & \cdots & \hat{\mathbf{x}}_{1n} \\ \hat{\mathbf{x}}_{21} & \hat{\mathbf{x}}_{22} & \cdots & \hat{\mathbf{x}}_{2n} \\ & & \ddots & \\ \hat{\mathbf{x}}_{m1} & \hat{\mathbf{x}}_{m2} & \cdots & \hat{\mathbf{x}}_{mn} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_m \end{bmatrix} \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 & \cdots & \mathbf{X}_n \end{bmatrix}$$

points ($3 \times n$)

cameras
($2m \times 3$)

The measurement matrix $\mathbf{D} = \mathbf{MS}$ must have rank 3!

Factorizing Measurement Matrix

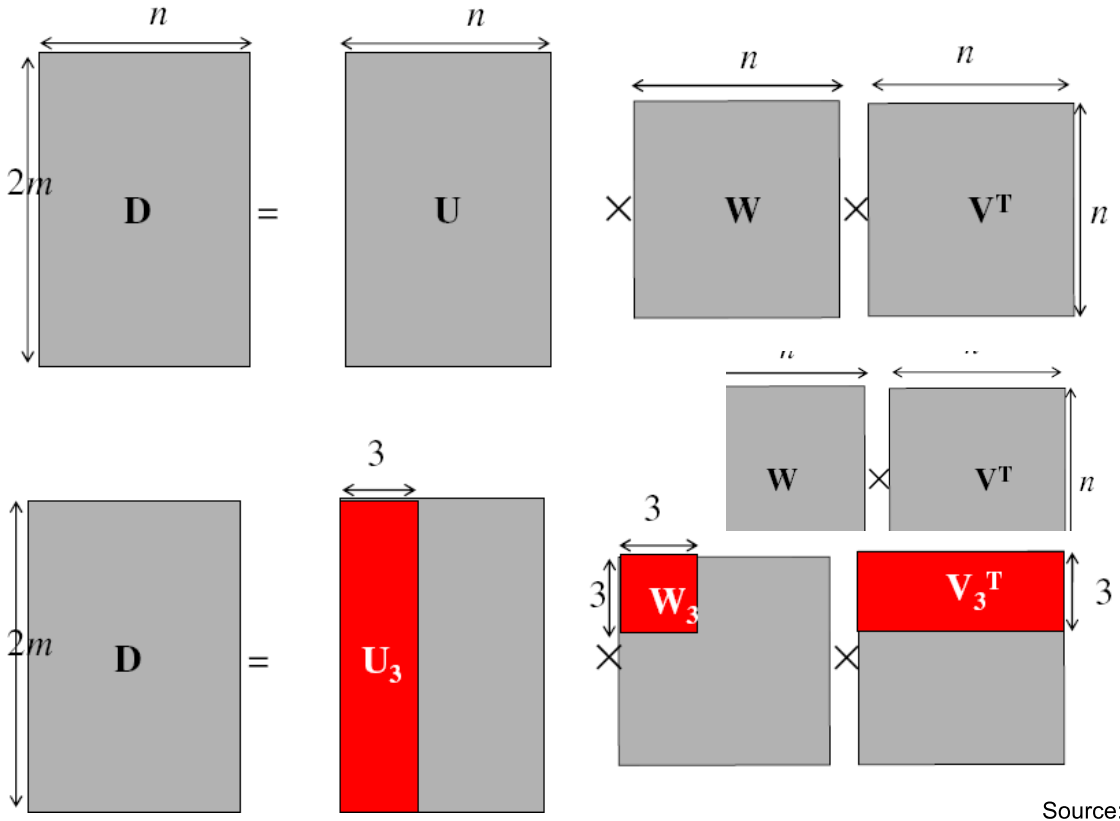


$$\mathbf{D} = \mathbf{MS}$$

Source: M. Hebert

Factorizing Measurement Matrix

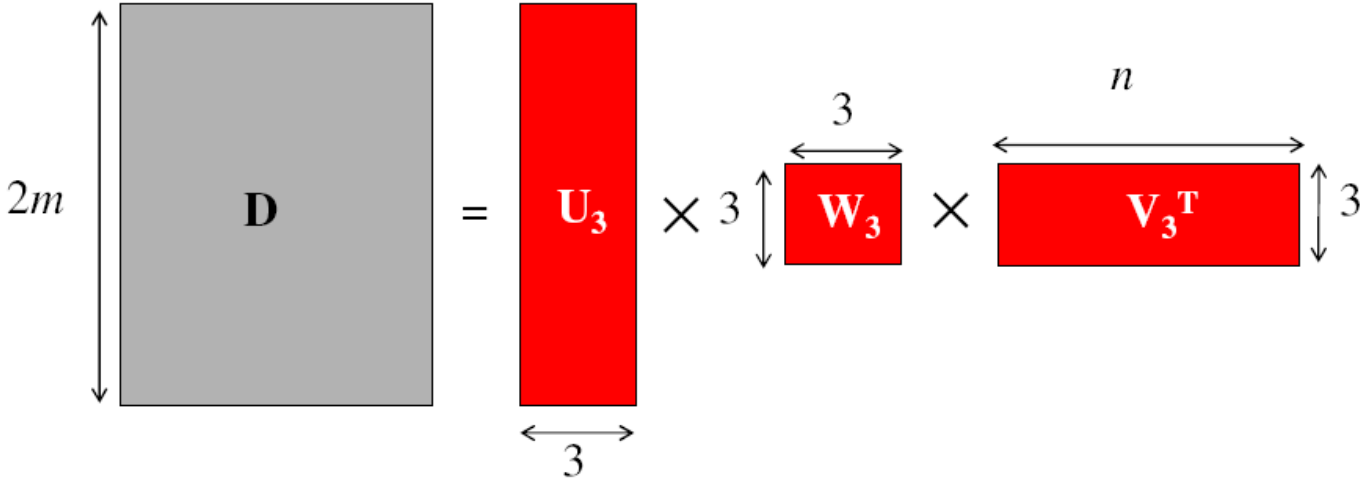
- Singular value decomposition of D:



Source: M. Hebert

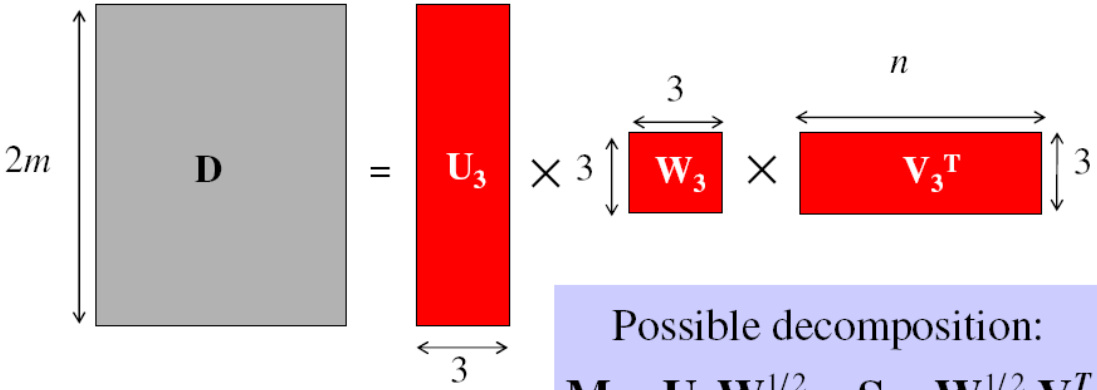
Factorizing Measurement Matrix

- Obtaining a factorization from SVD:



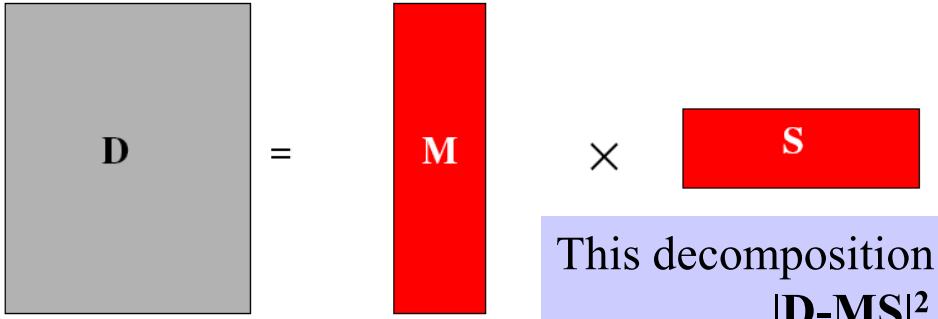
Factorizing Measurement Matrix

- Obtaining a factorization from SVD:



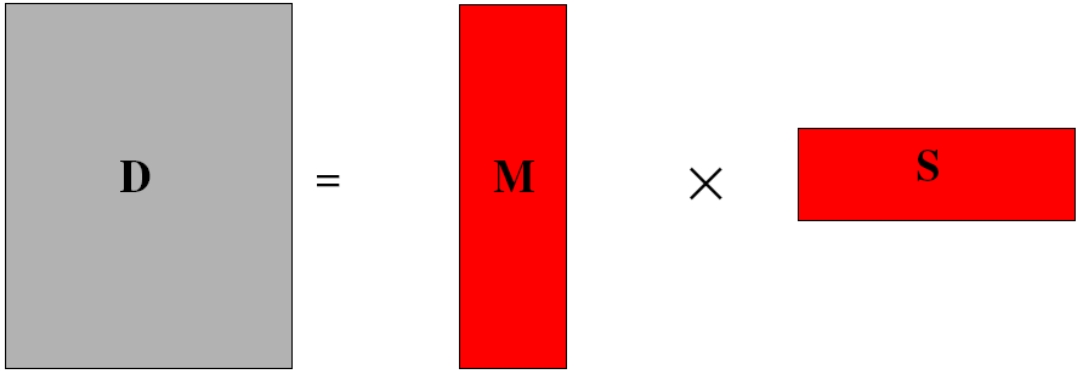
Possible decomposition:

$$\mathbf{M} = \mathbf{U}_3 \mathbf{W}_3^{1/2} \qquad \mathbf{S} = \mathbf{W}_3^{1/2} \mathbf{V}_3^T$$



This decomposition minimizes $|\mathbf{D}-\mathbf{MS}|^2$

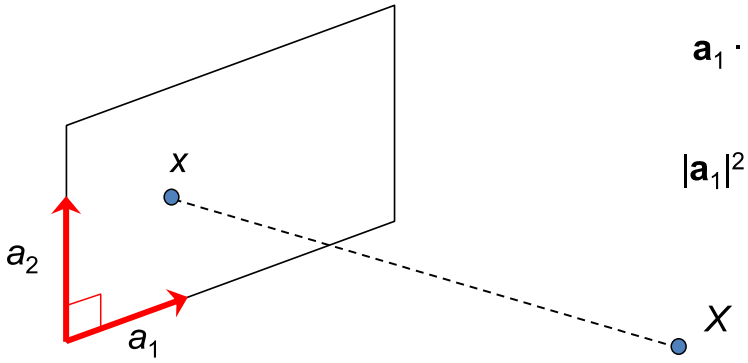
Affine Ambiguity



- The decomposition is not unique. We get the same **D** by using any 3×3 matrix **C** and applying the transformations **M** → **MC, S** →$\mathbf{C}^{-1}\mathbf{S}$
- That is because we have only an affine transformation and we have not enforced any Euclidean constraints (like forcing the image axes to be perpendicular, for example)

Eliminating Affine Ambiguity

- Orthographic: image axes are perpendicular and of unit length



$\mathbf{a}_1 \cdot \mathbf{a}_2 = 0$

$|\mathbf{a}_1|^2 = |\mathbf{a}_2|^2 = 1$

Eliminating Affine Ambiguity

Three equations for each image i

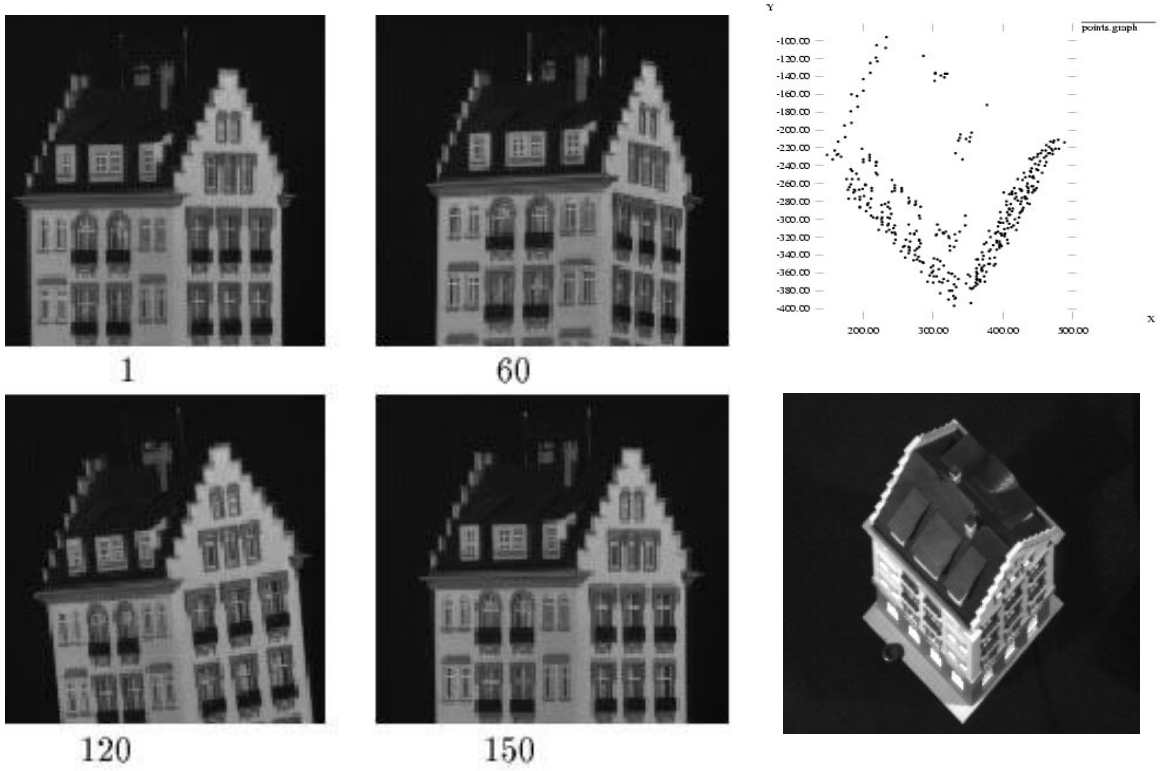
$$\begin{aligned}\tilde{\mathbf{a}}_{i1}^T \mathbf{C}\mathbf{C}^T \tilde{\mathbf{a}}_{i1}^T &= 1 \\ \tilde{\mathbf{a}}_{i2}^T \mathbf{C}\mathbf{C}^T \tilde{\mathbf{a}}_{i2}^T &= 1 \\ \tilde{\mathbf{a}}_{i1}^T \mathbf{C}\mathbf{C}^T \tilde{\mathbf{a}}_{i2}^T &= 0\end{aligned} \quad \text{where} \quad \tilde{\mathbf{A}}_i = \begin{bmatrix} \tilde{\mathbf{a}}_{i1}^T \\ \tilde{\mathbf{a}}_{i2}^T \end{bmatrix}$$

- Solve for $\mathbf{L} = \mathbf{C}\mathbf{C}^T$
- Recover **C** from **L** by Cholesky decomposition: $\mathbf{L} = \mathbf{C}\mathbf{C}^T$
- Update **A** and **X**: $\mathbf{A} = \tilde{\mathbf{A}}\mathbf{C}$, $\mathbf{X} = \mathbf{C}^{-1}\tilde{\mathbf{X}}$

Affine Structure from Motion Algorithm

- Given: m images and n features $\mathbf{x}_{ij}$
- For each image i, center the feature coordinates
- Construct a $2m \times n$ measurement matrix $\mathbf{D}$:
  - Column j contains the projection of point j in all views
  - Row i contains one coordinate of the projections of all the n points in image i
- Factorize $\mathbf{D}$:
  - Compute SVD: $\mathbf{D} = \mathbf{U}\,\mathbf{W}\,\mathbf{V}^{\mathbf{T}}$
  - Create $\mathbf{U}_3$ by taking the first 3 columns of $\mathbf{U}$
  - Create $\mathbf{V}_3$ by taking the first 3 columns of $\mathbf{V}$
  - Create $\mathbf{W}_3$ by taking the upper left 3×3 block of $\mathbf{W}$
- Create the motion and shape matrices:
  - $\mathbf{M} = \mathbf{U}_3\mathbf{W}_3^{1/2}$ and $\mathbf{S} = \mathbf{W}_3^{1/2}\,\mathbf{V}_3^{\mathrm{T}}$ (**or** $\mathbf{M} = \mathbf{U}_3$ and $\mathbf{S} = \mathbf{W}_3\mathbf{V}_3^{\mathrm{T}}$)
- Eliminate affine ambiguity

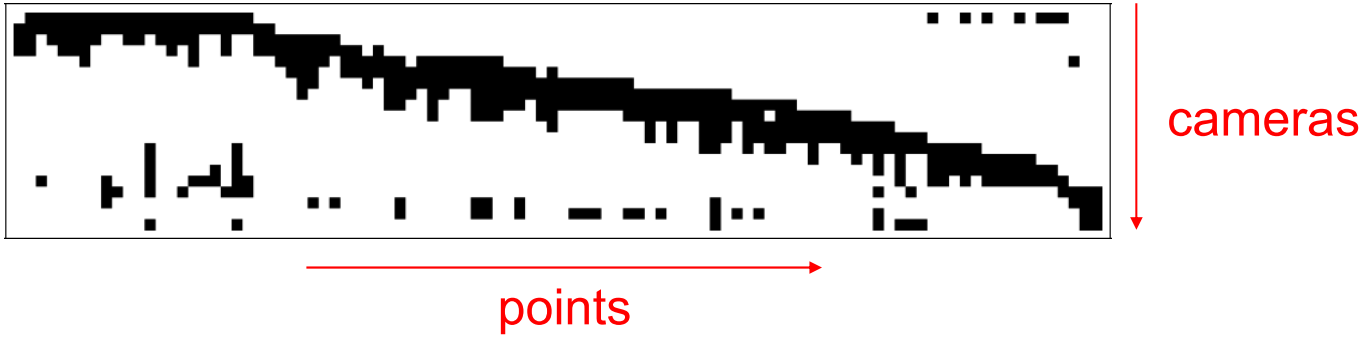
Affine Structure from Motion Results



C. Tomasi and T. Kanade. Shape and motion from image streams under orthography: A factorization method. *IJCV*, 9(2):137-154, November 1992.

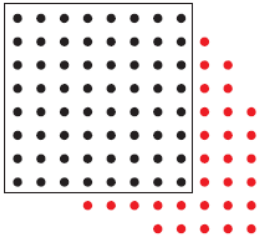
Dealing with Missing Data

- So far, we have assumed that all points are visible in all views
- In reality, the measurement matrix typically looks something like this:

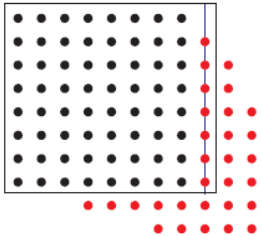


Dealing with Missing Data

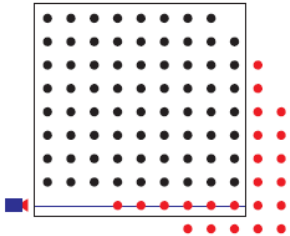
- Possible solution: decompose matrix into dense sub-blocks, factorize each sub-block, and fuse the results
  - Finding dense maximal sub-blocks of the matrix is NP-complete (equivalent to finding maximal cliques in a graph)
- Incremental bilinear refinement



(1) Perform factorization on a dense sub-block



(2) Solve for a new 3D point visible by at least two known cameras (linear least squares)



(3) Solve for a new camera that sees at least three known 3D points (linear least squares)

F. Rothganger, S. Lazebnik, C. Schmid, and J. Ponce. Segmenting, Modeling, and Matching Video Clips Containing Multiple Moving Objects. PAMI 2007.

Projective Structure from Motion

- Given: *m* images of *n* fixed 3D points

$$\mathbf{x}_{ij} = \frac{1}{z_{ij}} \mathbf{P}_i \mathbf{X}_j \ , \quad i = 1, \dots , m, \quad j = 1, \dots , n$$

$$z_{ij} = \boldsymbol{p}_3^T \mathbf{X}_j \qquad \textit{Where, } \mathbf{P}_i = \begin{bmatrix} \boldsymbol{p}_1^T \\ \boldsymbol{p}_2^T \\ \boldsymbol{p}_3^T \end{bmatrix}$$

$\boldsymbol{p}_3^T$ is the third row of the Projection matrix $\mathbf{P}_i$ with dim 1x4 and thus z_{ij} turns out to be the scale factor in the homogonous coordinates and $\mathbf{x}_{ij}$ is the 3x1 image coordinate $[u_{ij} \quad v_{ij} \quad 1]$.

So, we can write as: $z_{ij} \mathbf{x}_{ij} = \mathbf{P}_i \mathbf{X}_j , \quad i = 1, \dots , m, \quad j = 1, \dots , n$

Projective Structure from Motion

- Given: *m* images of *n* fixed 3D points

$$z_{ij}\,\mathbf{x}_{ij} = \mathbf{P}_i\,\mathbf{X}_j, \quad i = 1, \ldots, m, \quad j = 1, \ldots, n$$

- Problem: estimate *m* projection matrices $\mathbf{P}_i$ and *n* 3D points $\mathbf{X}_j$ from the *mn* correspondences $\mathbf{x}_{ij}$
- With no calibration info, cameras and points can only be recovered up to a 4x4 projective transformation **Q**:

$$\mathbf{X} \rightarrow \mathbf{QX}, \ \mathbf{P} \rightarrow \mathbf{PQ}^{-1}$$

- We can solve for structure and motion when

$$2mn >= 11m + 3n - 15$$

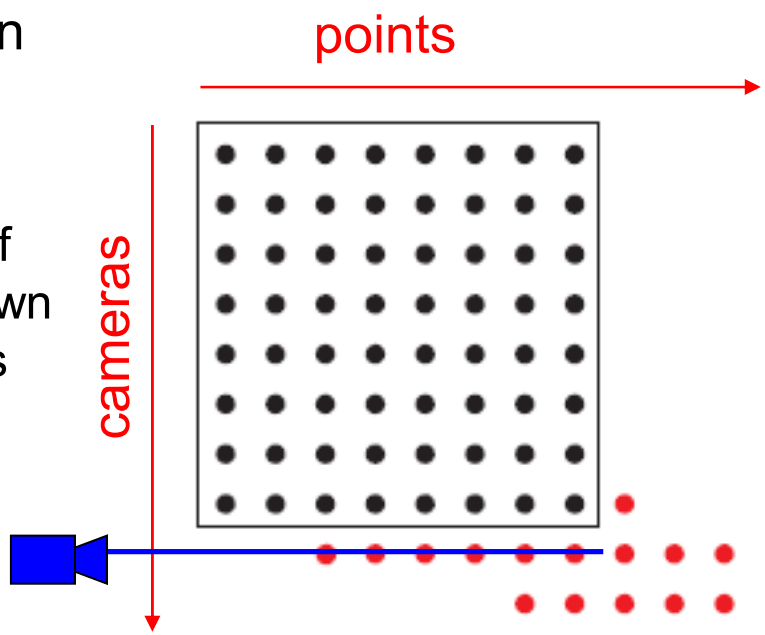
- For two cameras, at least 7 points are needed

Sequential Structure from Motion

- Initialize motion from two images using fundamental matrix

- Initialize structure by triangulation

- For each additional view:
  - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*



Initialize with Two Cameras

- Compute fundamental matrix $\mathbf{F}$ between the two views
- First camera matrix: $[\mathbf{I}|\mathbf{0}]$
- Second camera matrix: $[\mathbf{A}|\mathbf{b}]$
- Then $\mathbf{b}$ is the epipole ($\mathbf{F}^{\mathrm{T}}\mathbf{b} = 0$), $\mathbf{A} = -[\mathbf{b}_{\times}]\mathbf{F}$

Sequential Structure from Motion

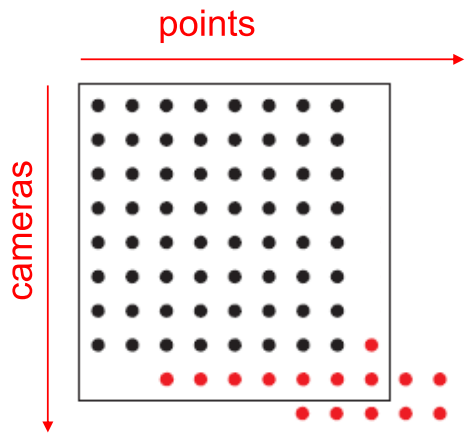
- Initialize motion from two images using fundamental matrix

- Initialize structure by triangulation

- For each additional view:
  - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*

  - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – *triangulation*

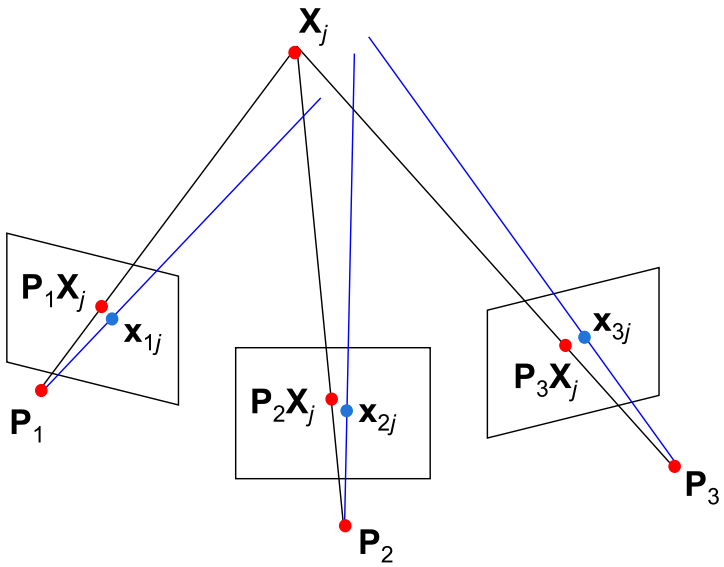
- Refine structure and motion: bundle adjustment



Bundle Adjustment

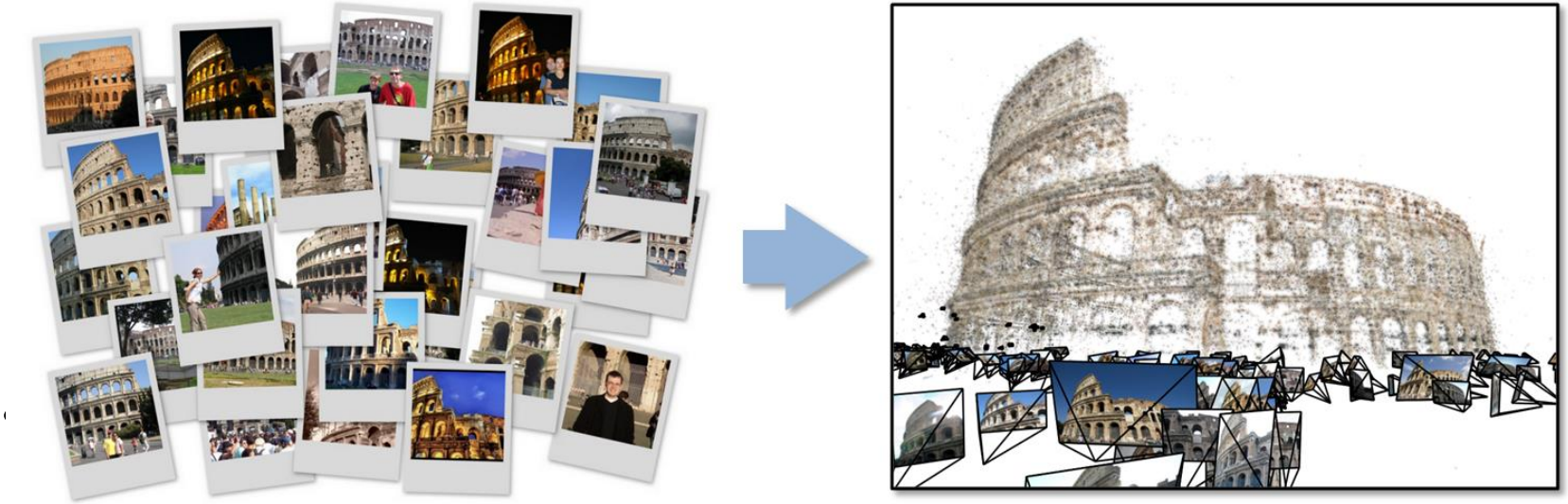
- Non-linear method for refining structure and motion
- Minimizing reprojection error

$$E(\mathbf{P},\mathbf{X}) = \sum_{i=1}^{m}\sum_{j=1}^{n} D\left(\mathbf{x}_{ij}, \mathbf{P}_i\mathbf{X}_j\right)^2$$

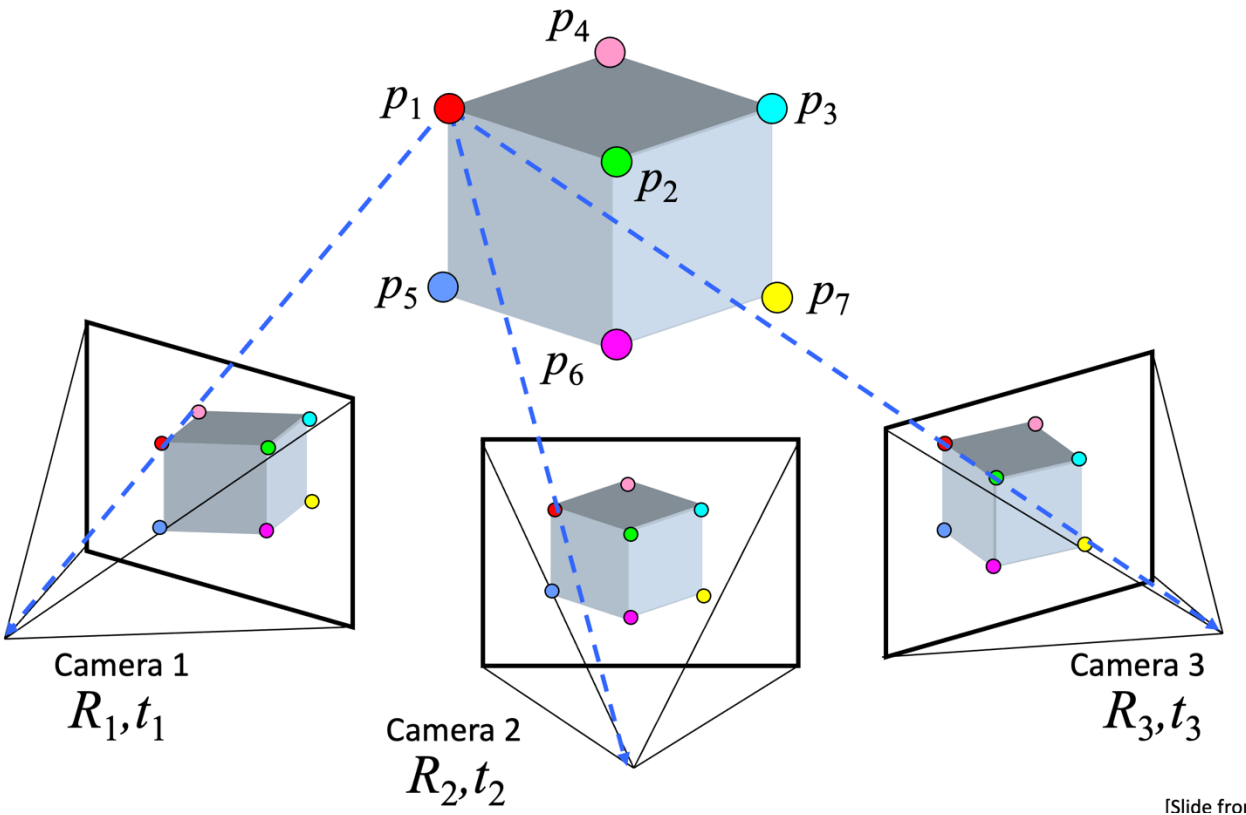


Large Scale SfM

- Recent work has built 3D models from large, unstructured online image collections
  - [Snavely06], [Li08], [Agarwal09], [Frahm10], Microsoft's PhotoSynth, …



Large Scale SfM



[Slide from

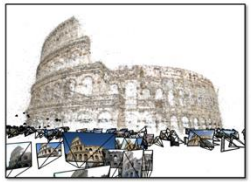
Large Scale SfM

Feature detection

Feature matching
Find scene points seen by multiple cameras

Initialization
Robustly estimate camera poses and/or scene points

Bundle adjustment
Refine camera poses **R, T** and scene structure **P**



Incremental Bundle Adjustment (IBA)

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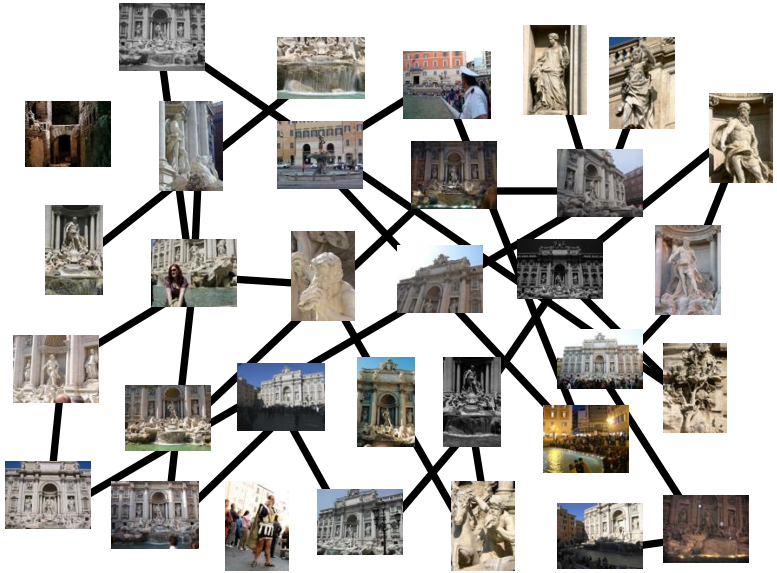
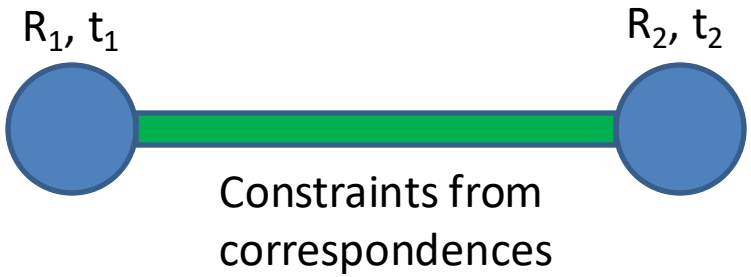
Large Scale SfM with IBA

- Image graph with pairwise point correspondences



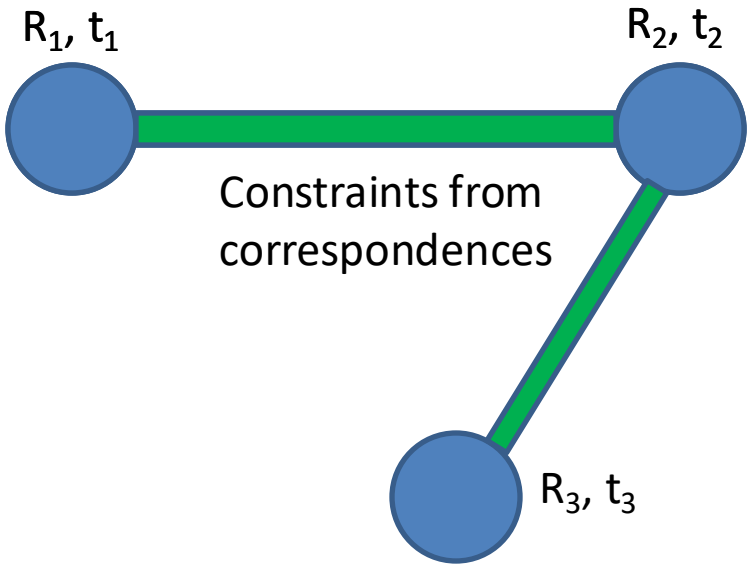
Large Scale SfM with IBA

Start with a seed pair



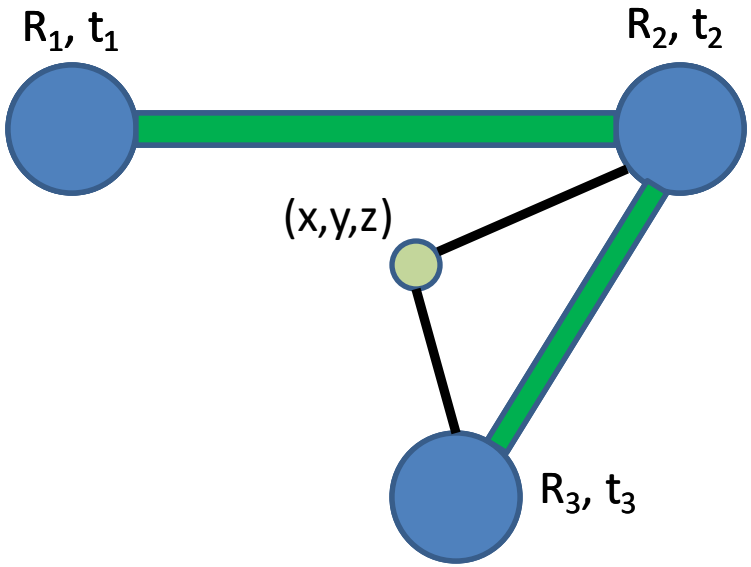
Large Scale SfM with IBA

Add new cameras to the seed



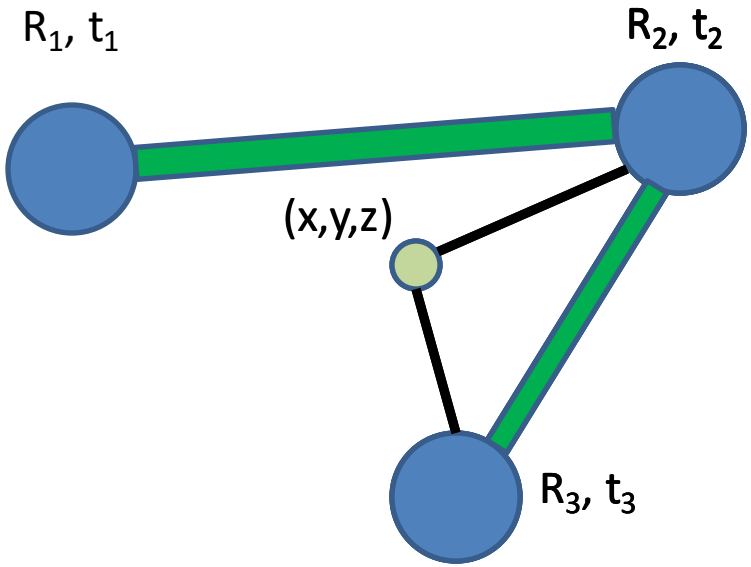
Large Scale SfM with IBA

Also calculate 3D positions of points that cameras see



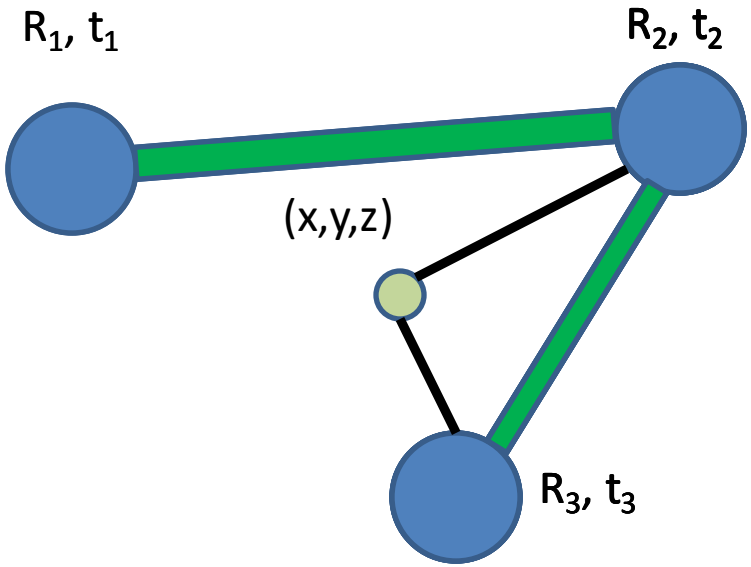
Large Scale SfM with IBA

Wiggle solution periodically to get a better solution



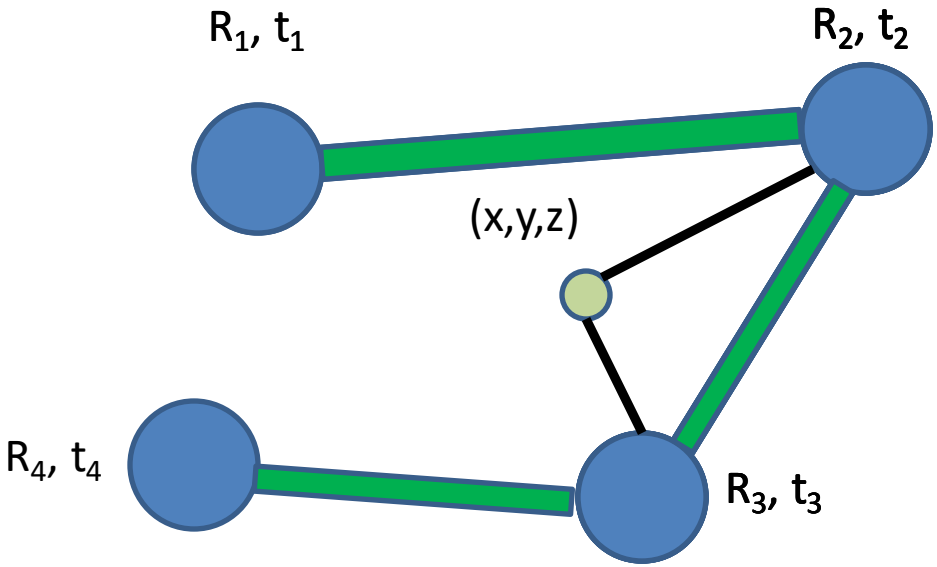
Large Scale SfM with IBA

Wiggle (Bundle adjust) solution periodically to get a better solution



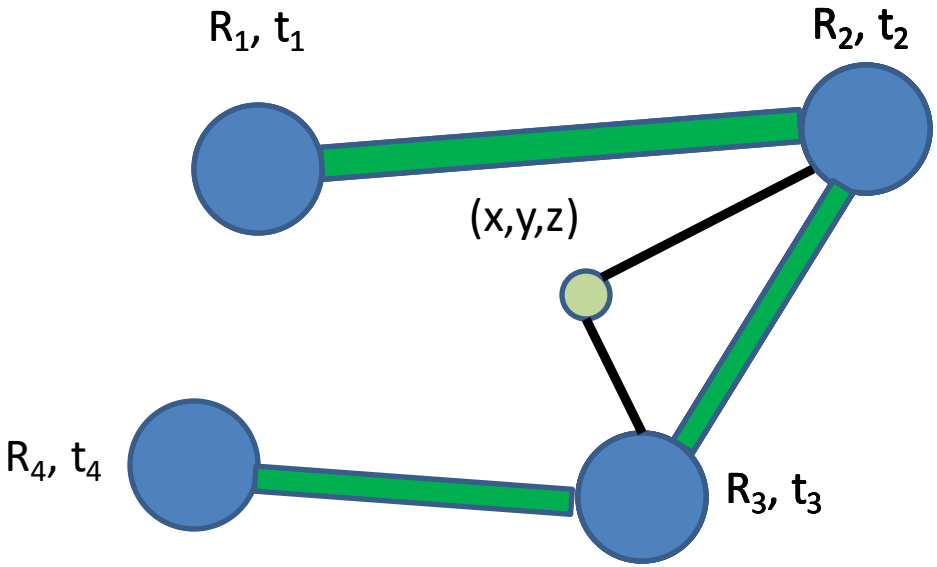
Large Scale SfM with IBA

Keep adding more cameras



Large Scale SfM with IBA

Keep adding more cameras

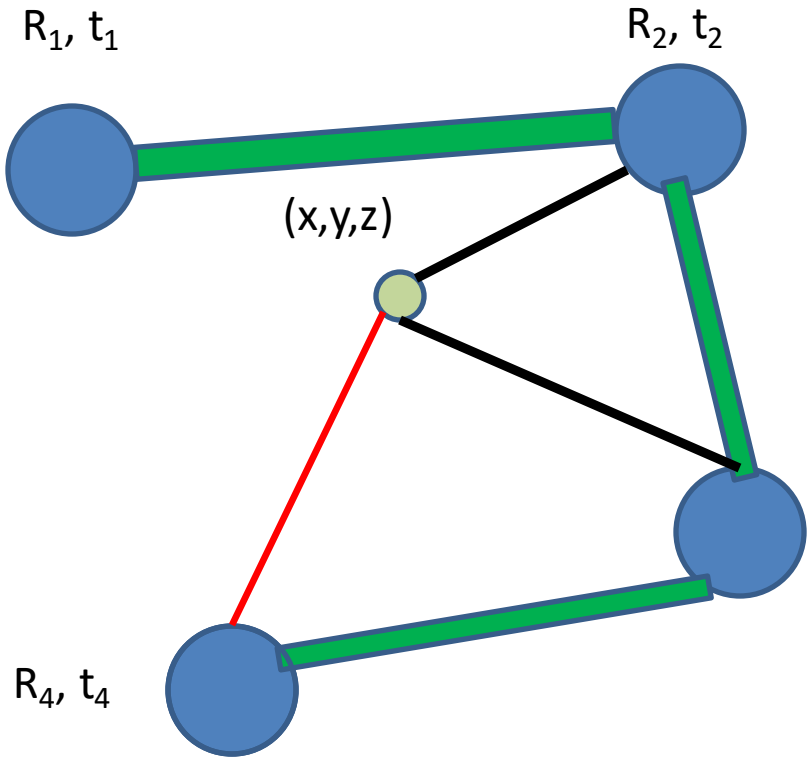


Challenges in Large Scale SfM with IBA

Incremental BA

- ✔ Works very well for many scenes
- ✘ Poor scalability, much use of bundle adjustment
- ✘ Poor results if a bad seed image set is chosen
- ✘ Drift and bad local minima for some scenes
- ✘ No way to add priors

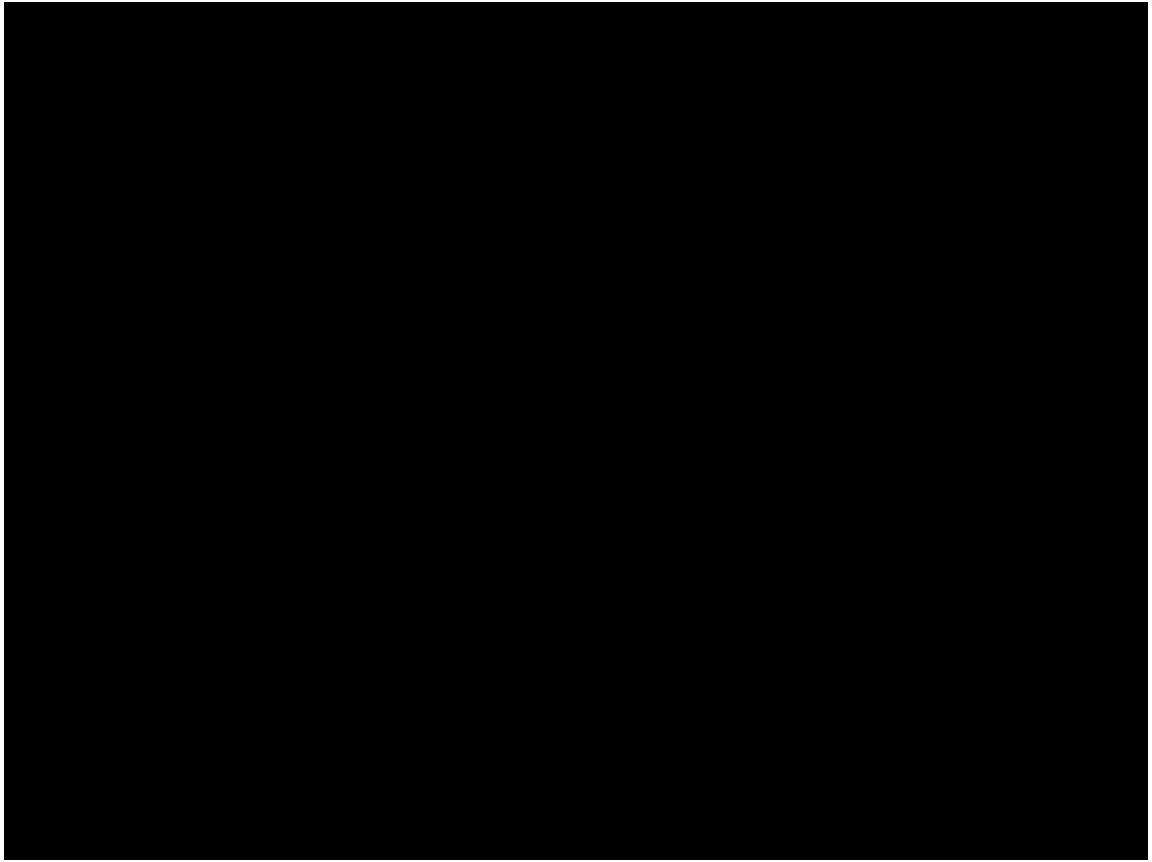
Large Scale SfM with Global Optimization



Solve for everything simultaneously

Discrete-Continuous Optimization for Large-scale Structure from Motion -David Crandall, Andrew Owens, Noah Snavely, Dan Huttenlocher

Large Scale SfM with Global Optimization



Thank You